

KAHBOD AEINI

Calgary, AB (Relocating to Waterloo, Jan 2026)
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SUMMARY

Graduate researcher in Computer Science with expertise in algorithms, distributed systems, and machine learning. Skilled in Python, C++, R, Haskell, and modern ML frameworks (PyTorch, TensorFlow), with experience in data engineering and large-scale system design. Seeking internship opportunities in software development, data/ML engineering, or distributed systems. Currently based in Calgary and relocating to Waterloo in Jan 2026.

EDUCATION

University of Waterloo PhD in Electrical and Computer Engineering (incoming)	Jan 2026 –
University of Calgary Master of Computer Science (GPA: 3.8/4.0)	2023 – 2025
Sharif University of Technology Bachelor of Computer Engineering	2019 – 2023

EXPERIENCE

Data Engineer, AI Lida	Feb 2022 – Sep 2022
- Built and optimized ETL pipelines using Python and PostgreSQL, improving data throughput and reliability. - Collaborated with the Security team to design secure data workflows for production systems. - Hired, trained, and mentored 5 data engineering interns.	
Research Assistant, University of Calgary	Jun 2024 – Present
- Designed and analyzed algorithms for concurrency and synchronization in distributed systems. - Proved space- and time-complexity bounds for randomized wait-free locks in shared-memory models. - Applied advanced probability and complexity theory to evaluate performance guarantees.	
Research Assistant, Johns Hopkins University (Remote)	Feb 2023 – Aug 2023
- Built ML models in Python/R to predict immune cell composition from lung cancer transcriptomics. - Applied feature selection, classification, and enrichment analysis to biological datasets.	

PUBLICATIONS

Kahbod Aeini, Dante Bencivenga, George Giakkoupis, Philipp Woelfel.
Simple and Efficient Randomized Wait-Free Lock.
Submitted to ACM-SIAM Symposium on Discrete Algorithms (SODA 2026), under review.

PROJECTS

Fraud Detection System (BSc Thesis)

GitHub

Developed and compared 3 machine learning models for detecting fraudulent banking transactions.

Secure Chat Application

GitHub

Built a Python-based secure messaging app with encryption features for safe communication.

Smart Water Supplier (IoT + Web App)

GitHub

Designed an IoT system with Raspberry Pi and Django for monitoring and managing water levels.

Information Retrieval System (NLP + SWD)

GitHub

Designed and implemented a retrieval system for twitter interaction and health articles with GUI, Classification and Clustering tweets and articles.

SKILLS

Programming: Python, C/C++, Java, R, Haskell, Prolog

ML/AI: PyTorch, TensorFlow, scikit-learn, NLP, GCNs

Data: PostgreSQL, MongoDB, MySQL, Pandas, NumPy

Tools: Git, Linux, Docker, NixOS

AWARDS

Departmental Research Award, University of Calgary (2023)

Top 100 National Universities Entrance Exam, Iran (2019)

National Elites Fellowship (2019)